

Does α -Amino- β -methylaminopropionic Acid (BMAA) Play a Role in Neurodegeneration?

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The association of α -amino- β -methylaminopropionic acid (BMAA) with elevated incidence of amyotrophic lateral sclerosis/Parkinson's disease complex (ALS/PDC) was first identified on the island of Guam. BMAA has been shown to be produced across the cyanobacterial order and its detection has been reported in a variety of aquatic and terrestrial environments worldwide, suggesting that it is ubiquitous. Various in vivo studies on rats, mice, chicks and monkeys have shown that it can cause neurodegenerative symptoms such as ataxia and convulsions. Zebrafish research has also shown disruption to neural development after BMAA exposure. In vitro studies on mice, rats and leeches have shown that BMAA acts predominantly on motor neurons. Observed increases in the generation of reactive oxygen species (ROS) and Ca^{2+} influx, coupled with disruption to mitochondrial activity and general neuronal death, indicate that the main mode of activity is via excitotoxic mechanisms. The current review pertaining to the neurotoxicity of BMAA clearly demonstrates its ability to adversely affect neural tissues, and implicates it as a potentially significant compound in the aetiology of neurodegenerative disease. When considering the potential adverse health effects upon exposure to this compound, further research to better understand the modes of toxicity of BMAA and the environmental exposure limits is essential.